

Stéfan OUMANSOUR

Computer graphics engineer

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I am a passionate and meticulous developer specializing in 3d rendering engines and low-level software architecture. I have a wide range of personal projects under my belt, as well as a year and a half of professional experience as an engine programmer at a small video game studio.

SKILLS

Languages: C++, C#, C, x86 Assembly, GLSL, HLSL, Python, Typescript, Javascript,
Technologies: Vulkan, DirectX 11, Metal, WebGL, OpenGL, Three.js, Unity, Unreal, Git, Linux
Fields: Game engines, graphics engines, Programming languages, Full-stack, Maths, Multi-threading, Low level programming, Real-time software
Spoken languages: French, English

PROFESSIONAL EXPERIENCES

Freelancing *January 2026 – current*

Freelance missions, primarily full-stack development (**VueJS**, **NestJS**, **Three.js**, **PHP Symfony**) :

3D printed trophy shop (VueJS, Three.js, Python Flask) :

- 3D trophy generation with topographical data downloaded from Copernicus
- Generate route mesh from GPS points
- Rendering and 3D model generation with **Three.js**
- Integration with Strava API
- Payments via Stripe

Game engine programmer, Fomenko *March 2024 – July 2025*

Developed a game engine and graphics engine in **C++**, **OpenGL**, **DirectX** et **Metal** at Fomenko on a small adventure game:

- PBR renderer, shadow maps, SSAO, water reflections
- Graphics abstraction layer
- Custom graphical user interface framework
- World editor
- Vegetation and mesh paint tools

PERSONAL PROJECTS

Vk-Engine, game engine *2024 - current*

Developed a game engine and graphics engine in **C++** and **Vulkan** with an entity system, an asset system and an editor:

- Forward clustered renderer
- PBR renderer, image based lighting, bloom, parallax occlusion mapping, shadow maps
- Skeletal animations
- Fast entity and components system
- ImGui editor with multi-viewport, scene management, material editor
- Jolt physics engine integration

Minecraft clone 2023

Recreated the Minecraft video-game in **C++** and **OpenGL**:

- Multi-noise and spline based terrain generation
- Chunk renderer with frustum culling
- Animations
- Terrain collision
- ImGui editor

Programming language 2024 - current

Worked on a compiled low-level systems programming language with manual memory management:

- Custom bytecode compilation
- Static typing
- Structs, pointers, arrays, functions with overloading
- Classic flow control structures (if/else, while and for loops, return, defer)
- Dynamic calls to C libraries
- Arbitrary compile time execution

Online pong game 2023

Full-stack web app in **Typescript** with **VueJS** and **NestJS**:

- **Websockets** based online pong game
- Two-factor authentication
- Real-time chat with public/private channels, user roles, direct messages
- Player profiles with match history, statistics and leaderboards
- Micro-service architecture with **NestJS** and a **PostgreSQL** database

FORMATION

42 school, Lyon, France 2021 - 2026

Expert en architecture informatique

Colbert High-School, Tourcoing, France 2017 - 2020

Baccalaureate (sciences and engineering)